

tion by heating the film.

The sculptured thin films that resulted from Dr Lakhtakia's research have a large range of uses including light filters, fluid concentration sensors, biosensors, electrical switching, lab-on-a-chip, bio-replication and bio-scaffolding (as a substrate for cell growth). Their use in the reproduction of biological surfaces such as the multifaceted eyes of house flies, for possible light-harvesting purposes, etc, is quite interesting.

Approach 2

"I'll keep studying nature; I'm sure it'll lead to great things!"

In a report published in the *National Geographic*, Dr Andrew Parker, a proponent of biomimetics and a research fellow at London's Natural History Museum and the University of Sydney, exclaimed that he could come up with 50 biomimetic project ideas merely by observing the preserved specimen in a museum. Indeed, when he saw a 45-million-year-old fly trapped in amber in a museum in Warsaw, Poland, he noticed that it had microscopic corrugations that reduced light reflection. This observation is now engineered into solar panels.

Parker is the kind of person who would walk along the coastline consciously looking at the environment, the natural phenomena and the creatures, searching for a spark to solve problems in engineering, materials science, medicine and other fields. Apparently, his study of antireflective coatings in moth eyes and iridescence in butterflies and beetles has led to an anti-counterfeiting technique and brighter screens for cellular phones, respectively.

So, in a very challenging approach some researchers such as Dr Parker keep studying nature, cataloguing the information and constantly calculating whether it can solve any related engineering problem. This requires a very keen mind, knowledge of many science and engineering disciplines, and a strong will—because hundreds of ideas and years of study might lead to just a few breakthroughs.

Self-repairing solar cells

In a very recent example of nature-inspired breakthroughs, a team of researchers at MIT, led by Prof. Michael Strano, has unveiled the logic for self-repairing solar cells, inspired by how plants perform photosynthesis. The design involves carbon nanotubes studded with phospholipid disks. The resulting photo-electrochemical solar cells are believed to be self-repairing and twice as efficient as the best solid-state solar panels available today.

Engineers try to prevent degradation in solar cells by using solid-state inorganic materials. Nature, on the other hand, does not try to prevent degradation, but uses various self-healing processes to anticipate and repair the damage caused to liquid-state organic materials. The process of photosynthesis includes mechanisms to periodically break down a plant's chemical-based engine into basic building blocks, renew the components and reassemble the engine. This process happens almost hourly.

Strano's design tries to mimic the refurbishing activity by using synthetic disk molecules called phospholipids, each of which carries its own internal reaction centre to convert light into an electric current. When dissolved in a solution containing carbon nanotubes, the disks self-assemble around them. Since carbon conducts electricity better than metals, the nanotubes enhance the transport of electrons freed inside the disks by their exposure to sunlight. Inside each disk there is a photosynthesis-like cyclical mechanism using seven different compounds that self-assemble into harvesters of light. By adding a surfactant, the seven-compound assembly can be broken apart into its original components. By forcing the whole solution through a filter that removes the surfactants, the original building blocks again self-assemble into a refurbished solar cell, much like the process followed by plants.

Inspired by the same process in plants, a team led by Jong Hyun Choi of Purdue University has also created artificial photosystems using optical nanomaterials to harvest solar energy that is converted to electrical power. This team's design exploits the unusual electrical properties of structures called single-wall carbon nanotubes, which they use as molecular wires in light-harvesting cells.

Approach 3

"I have a specific problem to solve; let me see if nature gives some ideas."

Dr Lakhtakia and an international team of researchers from USA, Italy and Spain were looking for a better strategy to channel sunlight into solar cells in small-scale installations. They believed that apart from harvesting direct sunlight, the cells should also recycle energy emitted by other lighting sources such as LED lamps. This requires solar cells that can harvest energy from diffused light. That is, we require solar cells with a large angular field of view to maximise the capture of incident light regardless of its direction.

Recently, the team consciously decided to look for solutions in nature. That is when they observed how difficult it was to catch houseflies.

"Houseflies have big eyes and can see 270 degrees around them. Certainly, this wide angular field of view arises from the positions of the two eyes; humans too would have a much

wider angular field of view than what they have (170 degrees) if their eyes were located on their temples. But, the team reasoned that the wide angular field of view could also arise from the compound eyes of the housefly *Musca domestica*," says Dr Lakhtakia.

Other insects such as the blowfly and the horsefly also have compound eyes. Basically, in these insects, each eye is made up of innumerable micro-sized cylindrical eyelets arrayed on a curved surface. Light propagating along the axis of an eyelet is collected to form an image, but light propagating in other directions and reaching an eyelet is absorbed by its dark side wall. Although the spatial resolution of the overall image formed in the brain by the fusion of the individual images is quite low, the field of view is very large.

Two years ago, the team started a two-phase research program to adapt the scalloped and curved outer surface of a compound eye to texture the exposed face of a solar-cell device. They report good progress in both phases. Although this success is a measure of